

Question:
At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

</tool_call> reasoning process here

At birth, the brain undergoes rapid development, but the regions develop at different rates. The brain stem, which controls vital functions like breathing and heart rate, is present and functional from birth. The thalamus, hypothalamus, and brain stem are more developed early on due to their roles in basic life-sustaining functions.

In contrast, the cerebral cortex, especially the prefrontal cortex (part of the frontal lobe), is the last to develop. This area is responsible for higher cognitive functions such as decision-making, reasoning, and planning. While the cortex is largely present at birth, the majority of its development, especially the complex synaptic connections and myelination, happens during the postnatal period, particularly in adolescence.

Therefore, the least developed part of the brain at birth – in terms of structural and functional maturity – is the frontal lobe, specifically the cortex, which continues to mature well into the second decade of life. The cortex is not fully myelinated or developed until later, making it the most underdeveloped in infancy compared to other structures.

So, the correct answer is the frontal lobe, although the cortex is functionally related, the frontal lobe is the region involved in early cognitive and executive functions and it is the last to mature.

Thus, the best choice considering developmental timelines and maturity is the frontal lobe.

< answer >C</answer >

